

VACAVILLE NUT TREE, CA, USA

WMO: 724828

Lat: 38.378N

Lon: 121.958W

Elev: 33

StdP: 100.93

Time Zone: -8.00 (NAP)

Period: 01-19

WBAN: 93241

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -0.8	(c) 1.0	(d) -10.2	(e) 1.6	(f) 14.3	(g) -7.8	(h) 1.9	(i) 14.0	(j) 9.8	(k) 10.6	(l) 8.5	(m) 11.3	(n) 0.2	(o) 350	(p) 0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	38.8	20.4	37.3	19.9	35.4	19.4	21.8	36.1	20.9	34.6	20.2	33.4	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	11.6	27.0	15.4	11.0	26.1	14.6	10.4	25.1	63.2	36.3	60.4	34.7	57.8	33.5	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.9	(b) 7.7	(c) 6.8	(e) -3.2	(f) 42.1	(g) 1.8	(h) 1.6	(i) -4.5	(j) 43.2	(k) -5.6	(l) 44.1	(m) -6.6	(n) 45.0	(o) -7.9	(p) 46.2		
			DB													
			WB													
				(e) -4.0	(f) 23.2	(g) 1.8	(h) 1.2	(i) -5.3	(j) 24.1	(k) -6.3	(l) 24.8	(m) -7.3	(n) 25.5	(o) -8.6	(p) 26.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	9.1	10.9	13.3	15.4	19.0	23.0	25.4	24.5	22.8	18.3	12.7	9.0
	DBStd	6.52	2.55	2.73	3.08	3.41	3.30	3.44	2.81	2.54	3.01	3.07	3.12	2.79
	HDD10.0	137	48	20	6	1	0	0	0	0	0	0	10	52
	HDD18.3	1288	287	208	157	100	31	4	0	0	2	38	171	290
	CDD10.0	2686	19	46	109	164	278	390	478	448	384	258	91	21
	CDD18.3	794	0	0	2	13	50	144	220	190	137	37	1	0
	CDH23.3	10973	1	5	67	256	793	2022	2925	2427	1831	608	39	0
(14) Wind	WSAvg	2.6	1.7	2.4	2.7	2.9	3.3	3.2	3.1	2.9	2.4	2.3	1.9	2.1
	PrecAvg	667	144	116	89	33	14	3	1	2	7	31	88	124
	PrecMax	1242	424	385	271	162	86	40	31	36	42	208	227	467
	PrecMin	207	2	2	0	1	0	0	0	0	0	0	0	7
	PrecStd	222	107	99	68	38	20	8	5	6	11	39	69	103
	DB	21.4	23.9	27.7	32.3	36.0	40.1	42.1	40.2	38.9	34.5	26.7	19.0	
	MCWB	10.8	14.2	16.0	17.5	19.5	21.3	21.8	20.2	19.0	17.4	14.9	11.2	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	18.7	21.8	25.2	29.0	32.6	37.7	39.0	37.6	37.0	31.4	24.1	17.3	
	MCWB	10.9	12.9	15.3	16.3	18.5	20.4	20.9	19.8	18.8	16.4	14.3	11.2	
	DB	16.5	19.0	22.8	26.4	30.5	35.2	37.2	35.9	34.7	28.9	21.7	15.9	
	MCWB	10.6	11.8	14.1	15.6	17.3	19.6	20.4	19.5	18.3	15.6	13.2	11.0	
	DB	14.4	17.2	20.9	23.6	27.8	32.8	34.9	33.7	32.5	27.1	19.0	14.1	
	MCWB	10.1	11.1	13.1	14.4	16.4	18.5	19.7	19.0	17.7	15.0	12.2	10.6	
	WB	14.5	15.8	17.4	19.4	21.3	23.0	23.0	21.9	21.2	19.2	17.0	15.2	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	16.9	20.7	25.2	29.2	32.6	36.5	39.1	36.4	34.7	29.8	23.6	16.8	
	WB	13.0	14.5	16.1	17.6	19.6	21.6	21.8	21.0	19.9	18.0	15.7	13.9	
	MCDB	15.9	19.0	23.6	26.9	31.3	35.3	36.9	34.7	33.3	28.3	20.7	15.2	
	WB	12.0	13.3	15.0	16.2	18.3	20.3	21.0	20.2	19.1	16.9	14.7	12.6	
	MCDB	14.0	17.3	21.6	24.5	28.8	33.3	35.4	33.6	32.2	26.1	19.1	14.0	
	WB	11.0	12.2	13.8	15.1	17.0	19.3	20.2	19.5	18.3	16.0	13.7	11.6	
	MCDB	13.6	15.6	19.1	22.6	26.5	31.5	33.8	32.4	31.0	24.3	17.8	13.1	
(35) Mean Daily Temperature Range	MDBR	10.2	11.2	11.9	13.0	14.6	16.7	18.4	18.2	17.8	15.3	12.4	9.6	
	5% DB	14.6	15.7	16.7	18.0	19.1	20.7	21.0	20.8	21.4	19.5	16.3	12.5	
	MCWBR	8.5	8.5	8.6	8.5	8.0	7.7	7.5	7.3	7.8	8.0	8.0	7.4	
	5% WB	9.9	12.2	15.2	16.6	17.9	19.4	19.9	19.2	19.1	16.5	12.3	7.2	
	MCWBR	6.6	7.3	8.3	8.5	8.3	7.9	7.5	7.2	7.5	7.5	6.9	5.2	
(40) Clear-Sky Solar Irradiance	taub	0.319	0.321	0.332	0.353	0.358	0.356	0.358	0.343	0.343	0.328	0.321	0.324	
	taud	2.525	2.524	2.480	2.402	2.414	2.425	2.411	2.444	2.452	2.519	2.524	2.513	
	Ebn at Noon	863	911	931	925	921	920	914	923	906	888	853	828	
	Edh at Noon	77	87	100	115	116	115	116	109	102	86	76	73	
(44) All-Sky Solar Radiation	RadAvg	2.23	3.24	4.61	6.10	7.33	8.14	8.04	7.18	5.88	4.17	2.73	1.98	
	RadStd	0.39	0.42	0.47	0.52	0.43	0.33	0.23	0.22	0.20	0.28	0.19	0.33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	+0.40	N/A	-1.13	N/A	N/A	N/A	N/A	-120	+136	+30

Nomenclature: See separate page